25900243 **Reg.No : .....**

Name : .....

MAHATMA GANDHI UNIVERSITY, KOTTAYAM

MGU-UGP (HONOURS) Regular EXAMINATION October 2025

Third SEMESTER

Discipline Specific Core Course (DSC) - MG3DSCBTS201 - GENERAL MICROBIOLOGY

(2024 ADMISSION ONWARDS)

Duration: 1 Hour 30 Minutes **Maximum Marks: 50**

Remember(K), Understand(U), Apply(A), Analyse(An), Evaluate(E), Create(C), Skill(S), Interest(I) and Appreciation(Ap)

Students should attempt at least one question from each course outcome to enhance their overall outcome attainability.

Part A

Fill in the blanks

Answer all questions

Each question carries **1** marks

1. Antonie van Leeuwenhoek, the first person to observe microorganisms under a microscope, described them as _____. [K] / [CO1]
2. Media that contain complex ingredients such as peptone, beef extract, or yeast extract are called _____ media. [U] / [CO2, CO3]
3. Viruses that infect bacteria are known as _____. [U] / [CO1]
4. Facultative anaerobes can grow in the presence or absence of _____. [U] / [CO3]
5. The infectious proteinaceous agents are called _____. [U] / [CO1]
6. The primary resource for the classification and identification of bacteria, often referred to as the 'bible of bacteriology', is _____. [K] / [CO1, CO2]
7. The enzyme responsible for transposition is _____. [U] / [CO5]
8. Bacteria capable of natural transformation are said to be _____. [U] / [CO5]
9. The enzyme _____ converts pyruvate to lactate in lactic acid fermentation. [U] / [CO3, CO5]

10. In the disc diffusion method, the zone of inhibition indicates the _____ of bacteria to the antibiotic. [U] / [CO3, CO4]

[1x10 = 10]

Part B

Short Answer Questions

Answer any **10** questions

Each question carries **2** marks

11. Differentiate between bacterial chromosome and plasmids. [U] / [CO5]

12. Name the four typical phases of a bacterial growth curve. [U] / [CO2, CO3]

13. Compare natural, semi-synthetic, and synthetic antibiotics. [U] / [CO3, CO4]

14. What is the principle of phase contrast microscopy? [K] / [CO1, CO2]

15. Describe the mechanism of action of sulfonamides. [U] / [CO3, CO4]

16. Differentiate between F^+ and F^- cells. [U] / [CO5]

17. What is a chemostat? [U] / [CO2, CO3]

18. State two factors that influence the effectiveness of a disinfectant. [U] / [CO3]

19. Name the three sugars present in TSI agar. [K] / [CO1, CO2]

20. How R plasmids confer resistance to bacteria. [U] / [CO5]

21. What is the chemical composition of peptidoglycan? [U] / [CO1]

22. Mention two nutritional factors essential for microbial growth. [K] / [CO3]

[2x10 = 20]

Part C

Short Essay Questions

Answer any **5** questions

Each question carries **4** marks

23. Define sterilization. Discuss in detail the various physical agents of sterilization with examples. [U] / [CO3]

24. Explain the steps involved in specialized transduction. [U] / [CO5]

25. Describe chemical agents of sterilization with examples. [U] / [CO3]

26. Write down the principle and procedure of Gram staining technique. [U] / [CO1, CO2]

27. Describe the various techniques for isolation of pure cultures. [U] / [CO2, CO3]

28. Write notes on transport media with examples. [U] / [CO3]

29. Write a short note on the industrial applications of yeast. [K] / [CO1]

[4x5 = 20]